

Muhammad Farhan

AI Engineer — LLMs & Agentic Systems | Generative AI | Python | FastAPI

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PROFESSIONAL SUMMARY

Computer Science graduate and self-driven AI Engineer specialising in LLMs and Agentic AI systems, with 1+ year of hands-on experience designing and deploying multi-agent architectures using LangChain and LangGraph. Built agentic workflows end to end — autonomous content-generation pipelines with evaluation gates, real-time conversational agents that assess human argument, and LLM pipelines that mine community discussion into validated product specifications — served behind production-style FastAPI endpoints. Placed **Top 5 in the GIKI Capstone Competition** (Advanced AI / Agentic AI Bootcamp) with Adalat AI. Grounded in classical ML and deep learning (CNNs, reinforcement learning, predictive modelling), giving depth beneath the agentic work rather than prompt-level familiarity alone.

TECHNICAL SKILLS

Generative AI	LLM Integration • Multi-Agent Systems • Agent Orchestration • Prompt Engineering • LangChain • LangGraph • Google Gemini • RAG
AI / ML	Predictive Modelling • Classification & Regression • Feature Engineering • Model Evaluation (Accuracy, Precision, Recall, F1, ROC-AUC) • Scikit-learn • Pandas • NumPy • Matplotlib
Deep Learning	CNNs • NLP • Neural Networks • Reinforcement Learning (PPO) • TensorFlow / Keras • PyTorch
Backend & APIs	FastAPI • REST API Design • Celery • Redis • Streamlit • Flask
Web	Next.js • React • TypeScript • Node.js • Express.js • JavaScript • Tailwind CSS
Data	PostgreSQL • SQLAlchemy • MongoDB • SQLite • Qdrant
Tools & Infra	Python (Advanced) • Git & GitHub • Docker • Jupyter • Vite • n8n
Languages	English (Professional) • Urdu (Native) • Pashto (Native)

FEATURED PROJECTS

AI Content Factory

Final Year Project (Team of 2) | 2025 – 2026 | Python, LangGraph, LangChain, LLMs, FastAPI, Next.js

- Built an orchestrator agent that takes a single topic, searches the web for authoritative sources, synthesises the research and produces a structured content outline for downstream agents.
- Added a blog generation agent that drafts the full article against that research and outline, keeping output factually consistent with the sources rather than free-generating from the prompt.
- Gated every draft behind an evaluation agent scoring factual accuracy, source reliability, structure, readability, originality and hallucination; failing drafts loop back for regeneration until they clear the threshold.
- Attached generated imagery to approved articles, then wired a repurposing pipeline producing a YouTube Short, podcast, video script, X post and LinkedIn post from each one.

Adalat AI — AI Moot Court Simulator

Capstone Project, GIKI | 2026 – Present | Python, LLMs, Multi-Agent Architecture, FastAPI

Achievement: Top 5 — GIKI Capstone Competition, Advanced AI / Agentic AI Bootcamp

github.com/farhan-hash404/MootCourtSimulator

- Built a courtroom simulation where a lawyer or law student takes the advocate role, presenting arguments, handling counterarguments and defending a position against an AI opposing counsel.
- Developed the opposing-counsel agent to argue the other side properly — raising relevant legal issues, challenging weak reasoning and responding in context rather than replying like a chatbot.
- Added an AI judge scoring legal reasoning, argument strength, evidence use and rebuttal handling against a defined rubric, returning a performance score with detailed feedback.
- Covered the full moot court cycle: case briefing, oral arguments, judicial questioning, rebuttals and final judgment, making practice repeatable without a human opponent.

Dev Signal — Pain-Point Discovery & PRD Platform

Full-Stack AI Platform | 2026 | Next.js, TypeScript, FastAPI, Google Gemini, PostgreSQL, Celery, Redis

github.com/farhan-hash404/Dev-Signal

- Built an automated discovery pipeline replacing manual forum research: async HTTPX scrapers pull top discussions across targeted subreddits (r/webdev, r/SaaS, r/programming) and the Stack Overflow API.
- Designed a two-pass Gemini pipeline — pass one extracts concrete pain points with severity, sentiment and frequency; pass two turns validated pain points into targeted SaaS product ideas.
- Generated full Product Requirement Documents per idea covering overview, problem statement, success metrics, tiered features, tech architecture, user stories and competitor analysis.
- Modelled the domain in SQLAlchemy 2.0 with UUID keys across analyses, pain points, ideas, PRDs, trends and competitors, backed by PostgreSQL with Alembic migrations.
- Served it behind FastAPI with Celery and Redis handling long-running scrape jobs, and a Next.js dashboard with client-side fallbacks so the UI stays available when the backend is unreachable.

Stick Fighter — In-Browser Reinforcement Learning

Personal Project | 2025 | TypeScript, HTML5 Canvas, Vite, PyTorch, Gymnasium

- Hand-rolled the entire neural network (Linear layers, manual backpropagation, Adam) with no TensorFlow.js, ONNX or WASM, shipping the game and agent in 51 KB with zero runtime dependencies.
- Implemented real PPO on an actor-critic MLP (46-dim observation, 2x64 tanh trunk, 15-action policy head plus value head) with GAE advantages, clipped surrogate objective, entropy bonus, gradient-norm clipping and target-KL early stopping, updating every 128 frames so learning is visible mid-fight.
- Solved cold start with a scripted aggression prior injected as a decaying additive logit bias that never enters the gradient, persisting learned weights to localStorage across sessions.
- Debugged a PPO correctness bug where behaviour and optimised policies diverged, pinning entropy at the uniform ceiling so nothing was learned despite healthy-looking metrics; the fix moved value loss 287 to 0.45 and approximate KL 0.46 to 0.003.
- Validated by a 42-check headless suite running the shipped code: with the scripted prior stripped, rounds won improved 11/12 to 12/12 and frames-to-win 619 to 258.

Kidney Disease Classification

Deep Learning / Computer Vision | 2025 | Python, TensorFlow / Keras, CNN, NumPy, Matplotlib

github.com/farhan-hash404/Kidney-Disease-Classification

- Trained a Convolutional Neural Network to classify kidney disease from medical scan images, covering the full pipeline from image preprocessing and augmentation through training and evaluation.
- Tuned architecture and hyperparameters, benchmarking with accuracy, precision, recall and confusion-matrix analysis to reduce false negatives on clinical-style data.

Wine Quality Analysis

Machine Learning / Data Analysis | Python, Pandas, Scikit-learn, Matplotlib

github.com/farhan-hash404/Wine-Quality-Analysis

- Performed exploratory analysis over physicochemical wine measurements, covering feature correlation, distribution analysis and outlier handling.
- Built and compared classification models for quality scoring, evaluating with accuracy, precision, recall and confusion-matrix analysis.

EDUCATION

BS Computer Science

University of Peshawar, Pakistan | 2022 – 2026

- All 8 semesters completed; degree awaiting official conferral. Available for full-time roles immediately.
- Relevant coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Databases, Software Engineering.
- Final Year Project: AI Content Factory — multi-agent content generation with an evaluation gate.

CERTIFICATIONS & TRAINING

- **Top 5 — GIKI Capstone Competition**, Advanced AI / Agentic AI Bootcamp (2026)
- AI / Agentic AI Capstone Program — GIKI (Ghulam Ishaq Khan Institute), 2026
- Fundamentals of Deep Learning — NVIDIA (Coursera)
- Fundamentals of Machine Learning — NVIDIA (Coursera)
- MERN Stack Development — NAVTTC (National Vocational & Technical Training Commission)
- Web and App Development — Saylani Mass IT Training (SMIT)

CAREER FOCUS

Actively pursuing AI/ML Engineer roles with a focus on Generative AI and Agentic AI systems. Planning postgraduate specialisation in GenAI and Agentic AI.